FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS – GRADE 10 | SUB-STRAND 1.1: REAL NUMBERS	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
This is your Final Explanation task. You will write a structured, multi-part response to the driving question using everything you have learned across all 8 lessons. Read the driving question carefully, then complete each section below in your own words. Use evidence from the Mathare North Health Centre phenomenon, class activities, and your model-building work to support every claim you make. Write in full sentences and show your mathematical reasoning clearly. Your goal is to demonstrate that you truly understand why some numbers "end neatly" and others go on forever — and why that difference matters in real life.

Section 1: Classify the Numbers from the Health Centre	
Look back at the numbers the health volunteer recorded: temperatures (36.8 °C, 37.0 °C, 37.5 °C), blood pressure readings (120/80, 135/90), shoe sizes (40, 41, 43), the average temperature you calculated, and the diagonal of the floor tile (√2). For each number, state whether it is rational or irrational and explain WHY using the definition of each set. Organise your answer clearly — you may use a table or bullet points.	The temperatures 36.8 °C, 37.0 °C, and 37.5 °C are all rational numbers because each one can be written as an exact fraction: 36.8 = 368/10, 37.0 = 37/1, and 37.5 = 375/10. The blood pressure readings such as 120/80 are already written as fractions of two integers, so they are rational. Shoe sizes 40, 41, and 43 are whole numbers, and every whole number is rational because you can write it over 1 (e.g. 41 = 41/1). The average temperature (36.8 + 37.0 + 37.5) ÷ 3 = 111.3 ÷ 3 = 37.1 °C, which equals 371/10 — still rational, and it terminates. However, the diagonal of the 1 m × 1 m floor tile is √2. Using the Pythagorean theorem: diagonal^2 = $1^2 + 1^2 = 2$, so diagonal = √2 ≈ 1.41421356… This decimal never ends and never repeats, which means it cannot be written as a fraction p/q where p and q are integers and q ≠ 0. Therefore √2 is irrational. It belongs to the real number system but sits outside the rational numbers.

Section 2: Explain the Real Number System	
Using a labelled diagram OR a	All the numbers the health volunteer used belong to the real number system (ℝ), which contains every number that can be placed

well-organised written explanation, describe how all the numbers you classified in Section 1 fit into the real number system. Your explanation must include: natural numbers, whole numbers, integers, rational numbers (terminating and recurring decimals), and irrational numbers. Show how these sets are nested inside one another and give at least one Kenyan everyday example for each set.	on a number line. Inside ℝ, we first split numbers into rational (ℚ) and irrational sets. Rational numbers include all values that can be written as p/q (p, q ∈ ℤ, q ≠ 0). Within rational numbers, we find integers (ℤ) — for example, the shoe size 43 or the number of patients seen, which could be −2 if we were recording a debt of 2 patients in a tally error. Inside integers are whole numbers (𝕎), such as the 0 patients in a ward, and inside whole numbers are natural numbers (ℕ) like 1, 2, 3 — for instance, the 3 temperature readings taken that morning. Rational decimals are either terminating (like 36.8 °C, which ends after one decimal place, similar to the price of ½ kg of ugali flour at Ksh 27.50) or recurring (like 1/3 = 0.333…, similar to splitting Ksh 10 equally among 3 children at a Kisumu market stall — each gets 3.333… shillings). Irrational numbers, like √2 (the tile diagonal) or π (used when calculating the area of a circular sufuria lid), cannot be expressed as fractions. They sit inside ℝ but completely outside ℚ. The sets are nested: ℕ ⊂ 𝕎 ⊂ ℤ ⊂ ℚ ⊂ ℝ, with irrational numbers filling the remaining space inside ℝ.

Section 3: Answer the Driving Question Directly	
Now answer the driving question in a focused paragraph of at least 8 sentences: Why do some numbers from our everyday Kenyan lives 'end neatly' while others go on forever — and how does knowing the difference help us make better decisions? Your answer must: (a) explain the mathematical reason some decimals terminate and others do not, (b) connect back to the Mathare North Health Centre phenomenon specifically, and (c) give at least ONE other Kenyan real-life context where this distinction matters.	Some numbers end neatly because they can be expressed as fractions whose denominators, when fully simplified, have only the prime factors 2 and/or 5. When you divide such a fraction, the long division process eventually reaches a remainder of zero, so the decimal stops — we call these terminating decimals. For example, the health volunteer's average temperature of 37.1 °C = 371/10, and since 10 = 2 × 5, the decimal terminates. Other rational numbers produce recurring decimals because their simplified denominators contain prime factors other than 2 or 5 — the remainder in long division cycles and never reaches zero. Irrational numbers like √2, however, are in a different category altogether: they cannot be written as any fraction of integers, so their decimal expansions go on forever without ever settling into a repeating pattern. At Mathare North Health Centre, the temperatures and blood pressure readings are rational because medical instruments are designed to record values to a fixed number of decimal places, making calculations — like averages — clean and trustworthy for diagnosis. The floor tile diagonal, on the other hand, involves √2, which is irrational; a builder tiling that waiting room cannot write its exact length, so they must use an approximation (e.g. 1.414 m) and accept a tiny rounding error. Knowing this difference matters in real decisions: a nurse who mistakes an irrational approximation for an exact value could miscalculate a medicine dosage, while a farmer in the Rift Valley using π to estimate how much irrigation pipe fits around a circular field must understand that π ≈ 3.142 is only an approximation and should build in a margin of error when purchasing materials. In short, understanding the nature of a number — rational or irrational — tells you whether you are working with an exact value or an approximation, and that knowledge directly affects the accuracy and safety of every calculation you make.

Section 4: Operations and Decisions with Real Numbers	
A maize farmer in Kericho wants to fence a square plot whose area is exactly 5 m². She calculates the side length as √5	(a) √5 is irrational. The number 5 is not a perfect square (no integer multiplied by itself gives 5), so √5 cannot be simplified to a fraction. On a calculator, √5 ≈ 2.2360679…, a decimal that never terminates or repeats, confirming it is irrational. (b) When you add an irrational number to a non-zero rational number, the result is always irrational. Here, √5 + 3.2 = 2.2360679… + 3.2 = 5.2360679…, which still never terminates or repeats. If it were rational, we could write √5 = (√5 + 3.2) − 3.2, which would mean

m and then tries to add it to a rational measurement of 3.2 m (the width of her gate). Show and explain: (a) what type of number √5 is and how you know, (b) what happens when you add √5 + 3.2 — is the result rational or irrational? — and (c) what practical advice you would give the farmer when she goes to buy fencing wire, based on your understanding of rational and irrational numbers.	irrational = rational − rational = rational — a contradiction. So the sum is irrational. (c) Because √5 is irrational, the farmer cannot buy an exact whole-number or simple-fraction length of fencing wire for one side. I would advise her to round √5 up to the next sensible approximation — say 2.24 m — rather than rounding down, so she does not end up short of wire. She should then multiply 4 × 2.24 = 8.96 m for the four sides and add her gate width of 3.2 m, giving a total of about 12.16 m. When purchasing, she should buy at least 12.2 m to account for the rounding approximation. Understanding that √5 is irrational — and therefore can only be approximated — is what tells her she needs a safety margin; a farmer who thinks √5 is exactly 2.2 could end up 0.14 m short per side, leaving her fence incomplete.

Section 5: Reflect on Your Model	
Look back at the number-line model or concept map you built and revised across all 8 lessons. (a) Describe ONE important change you made to your model during the lessons and explain what new understanding caused that change. (b) Identify ONE thing you found genuinely surprising or difficult about real numbers during this investigation. (c) Write one new question you still have about real numbers or about how they are used in Kenyan science, health, or business contexts.	(a) At the start of the investigation, I placed √2 on my number line with a question mark because I was not sure it could even live on the number line — I thought 'forever decimals' might be too big or too messy to have a fixed position. After Lesson 4, when we used a compass-and-ruler construction to mark √2 exactly as the hypotenuse of a right-angled triangle with both legs equal to 1, I revised my model to show √2 sitting precisely between 1.4 and 1.5 on the number line. This changed my understanding: irrational numbers are not vague or approximate — they have exact positions on the real number line; it is only their decimal form that is infinite. (b) I found it genuinely surprising that there are infinitely more irrational numbers than rational numbers on the number line, even though in everyday Kenyan life — prices at Gikomba Market, weights at a Mombasa fish stall, distances on a matatu route — almost every number we use is rational. It seemed backwards that the 'weird' numbers would actually be more common. (c) My new question is: when doctors at Kenyatta National Hospital use formulas to calculate a patient's Body Mass Index (BMI) or kidney filtration rate, do any of those formulas ever produce truly irrational outputs, or does rounding always make them rational in practice — and does that rounding ever cause medical errors?

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Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Classification & Definition of Real Number Sets	Correctly classifies all numbers from the phenomenon as rational or irrational with precise mathematical justification (p/q definition, prime factorisation of denominators, or proof by contradiction). Accurately maps all sets ($\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset$	Correctly classifies most numbers with adequate justification. Describes the nested structure of real number sets with minor gaps or omissions in Kenyan examples.	Attempts to classify numbers but makes errors (e.g. confuses recurring decimals with irrational numbers, or misplaces integers). Shows partial understanding of the real number system structure.

	$\mathbb{Q} \subset \mathbb{R}$) with relevant Kenyan examples for each.		
Explanation of Terminating vs. Non-Terminating Decimals	Clearly and accurately explains why some decimals terminate (denominator's prime factors are only 2 and/or 5) and why others recur or go on forever. Connects this reasoning directly and explicitly to the Mathare North phenomenon and at least one other Kenyan context.	Explains the terminating/non-terminating distinction using correct mathematical ideas, with a clear link to the phenomenon. The explanation of the underlying reason (prime factors of denominator) may be partially developed.	Identifies that some decimals end and others do not, but the explanation relies mainly on calculator observations rather than mathematical reasoning. Connection to the phenomenon is weak or missing.
Mathematical Reasoning & Operations with Real Numbers	Correctly performs and explains operations involving rational and irrational numbers (e.g. rational + irrational = irrational). Uses logical reasoning or proof by contradiction to justify conclusions. Work is shown clearly with no computational errors.	Mostly correct operations and reasoning with minor errors. Justifies conclusions using mathematical logic, though the argument may lack full rigour (e.g. states the rule without proving it).	Attempts operations but makes errors in computation or reasoning. May state correct answers without mathematical justification, or confuse the result type (e.g. thinks rational + irrational could be rational).
Real-World Application & Decision-Making	Provides detailed, specific, and mathematically sound advice for real-world Kenyan scenarios (health, farming, construction, etc.). Clearly explains how knowing whether a number is rational or irrational directly changes the decision or introduces a need for a safety margin or approximation strategy.	Gives relevant real-world advice with a clear link to rational/irrational number properties. The practical implication is correct but may lack specific numerical detail or depth of explanation.	Attempts to apply knowledge to a real-world context but the connection to rational vs. irrational numbers is vague or incorrect. Advice given may not follow logically from the mathematics.
Model Reflection & Mathematical Growth	Articulates a specific, meaningful change to their model with a clear explanation of the new conceptual understanding that drove the change. Identifies a genuinely substantive surprise or difficulty. Poses a thoughtful, mathematically curious follow-up question rooted in a Kenyan context.	Describes a valid model change and links it to a lesson activity. Identifies a difficulty or surprise. Follow-up question is relevant but may be surface-level rather than deeply investigative.	Reflection is vague or describes only superficial changes (e.g. 'I added more numbers'). Difficulty or surprise stated without mathematical explanation. Follow-up question is missing or unrelated to real numbers.